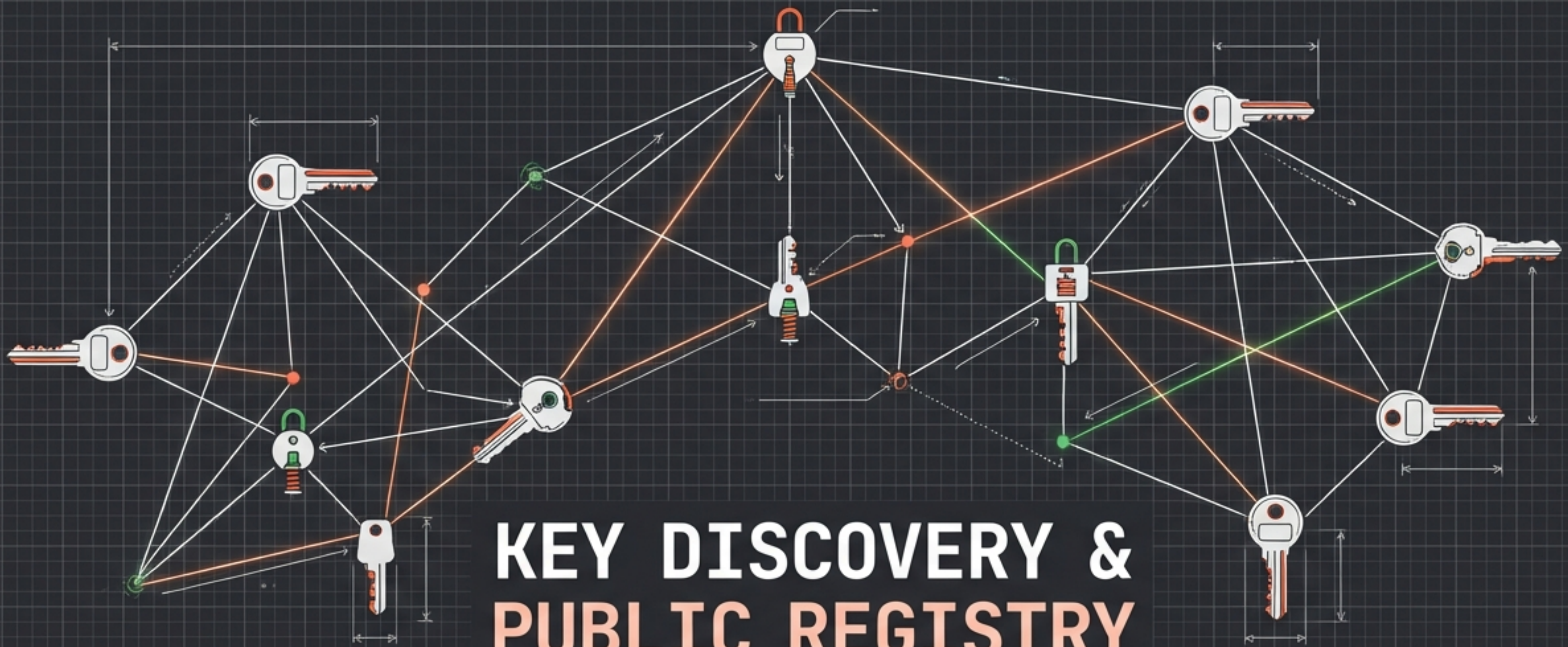
PROJECT: KEY DISCOVERY

VERSION: v0.4.12



KEY DISCOVERY & PUBLIC REGISTRY

Developer Brief: From manual PEM blobs to seamless discovery.

DATE: 20 FEB 2026
CLASSIFICATION: B3 AS B1 / D1
STATUS: CURRENT PRIORITY

PROGRESS TRACKER: ADMIN PKI & DISCOVERY

LENGTH: 50% | STATUS: COMPLETE

50%

20%

DONE: ADMIN PKI (24H BUILD)

50%

- Key generation & Export/Import active
- Encrypt/Decrypt production ready
- Deployed in ~24 hours

LENGTH: 50% | STATUS: IN PROGRESS

50%

NEXT: DISCOVERY (CURRENT)

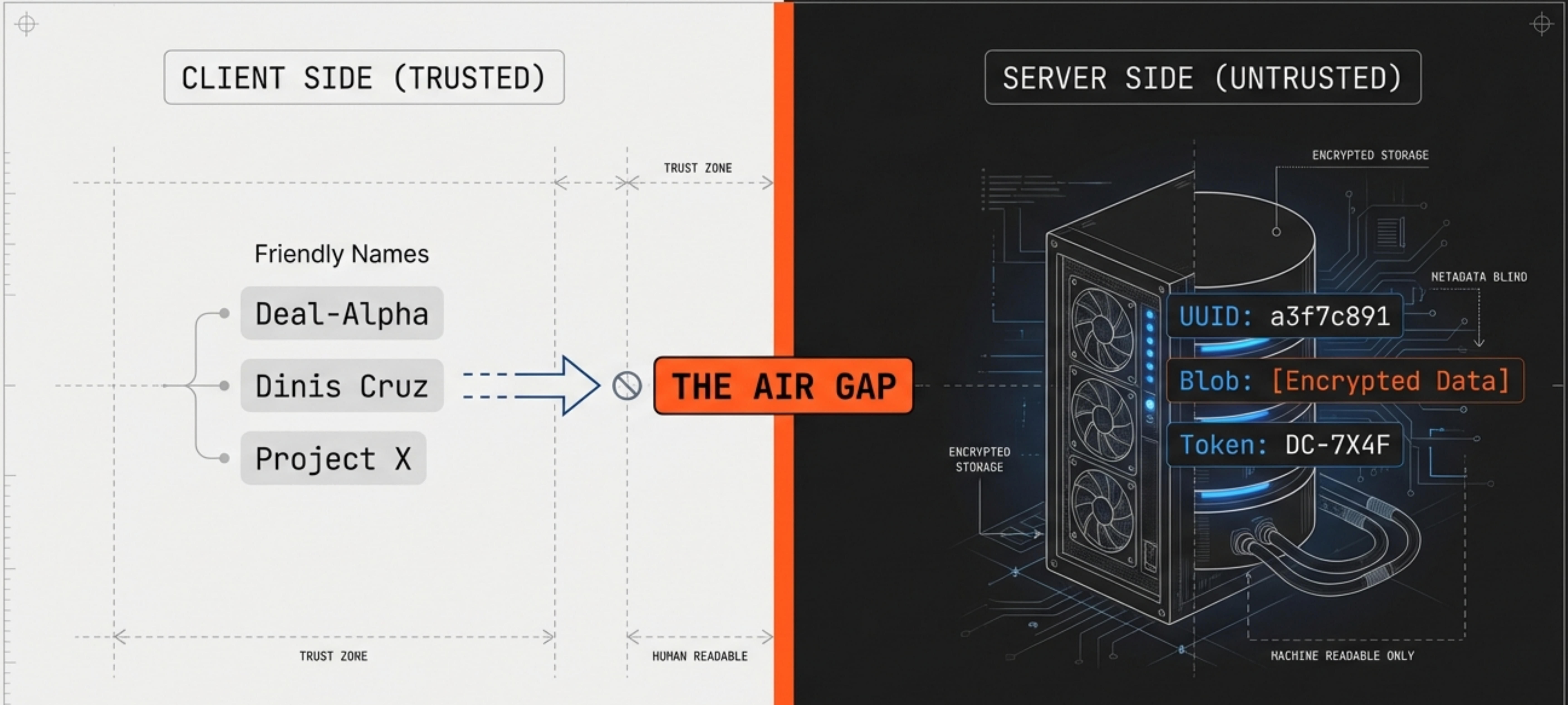
20%

50%

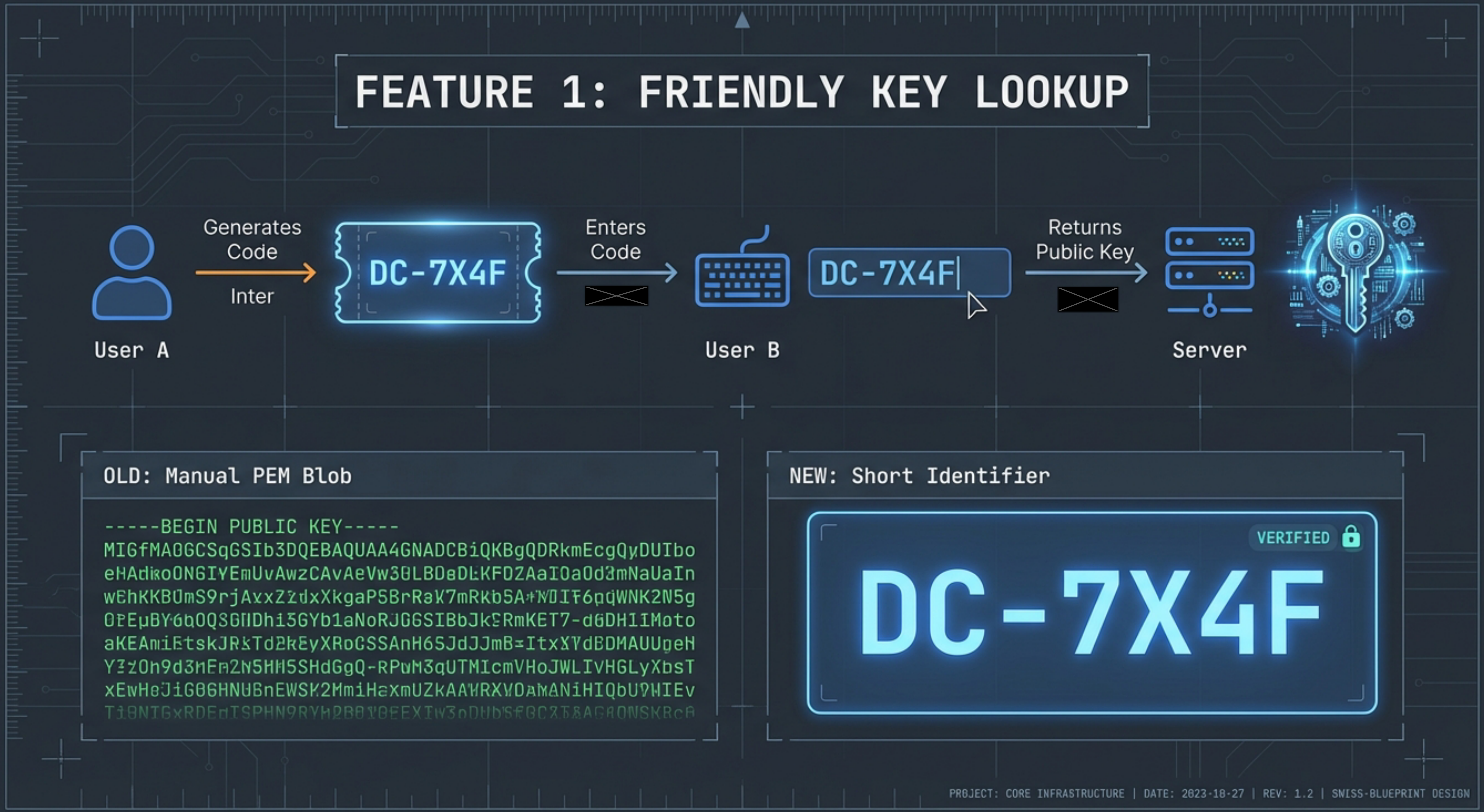
- Eliminate manual copy-paste friction
- Implement Friendly Key Lookup
- Build Public Registry

“We built the engine; now we build the steering wheel.”

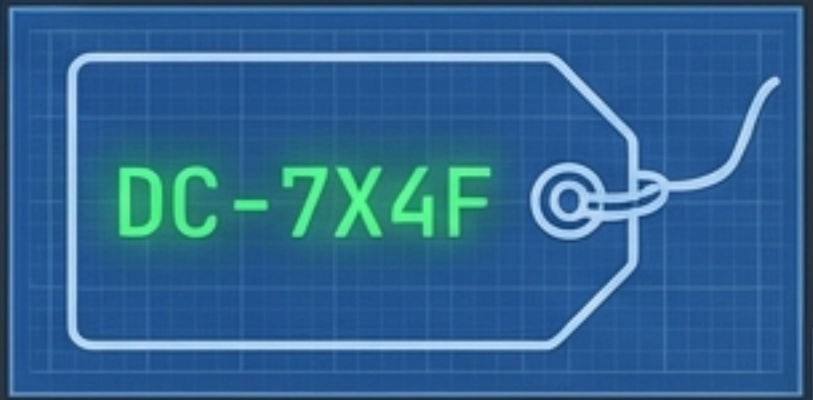
THE IRON RULE: ZERO PLAINTEXT ON SERVER



Security Rationale: Zero metadata leakage. No strings to inject. The server processes tokens, not meanings.


FEATURE 1: FRIENDLY KEY LOOKUP
Generates Code
Inter
User A
DC-7X4F
Enters Code
[redacted]
User B
DC-7X4F
Returns Public Key
[redacted]
Server
OLD: Manual PEM Blob
-----BEGIN PUBLIC KEY-----
MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDRkmEcgQyDUIbo
NEW: Short Identifier
VERIFIED
DC-7X4F
PROJECT: CORE INFRASTRUCTURE | DATE: 2023-10-27 | REV: 1.2 | SWISS-BLUEPRINT DESIGN

SOLVING THE NAMING PROBLEM

HASHING	RAW OBJ_ID	HYBRID (CHOSEN)
Server stores sha256(name).	Server stores UUID only.	Short Alphanumeric Token.
 SHREDDER	a3f7c891 BARCODE	 LUGGAGE TAG
REJECTED	REJECTED (Too complex)	Opaque to server (No metadata) Readable by humans Mapped to contacts in Client DB

The Hybrid approach balances strict security with usability.

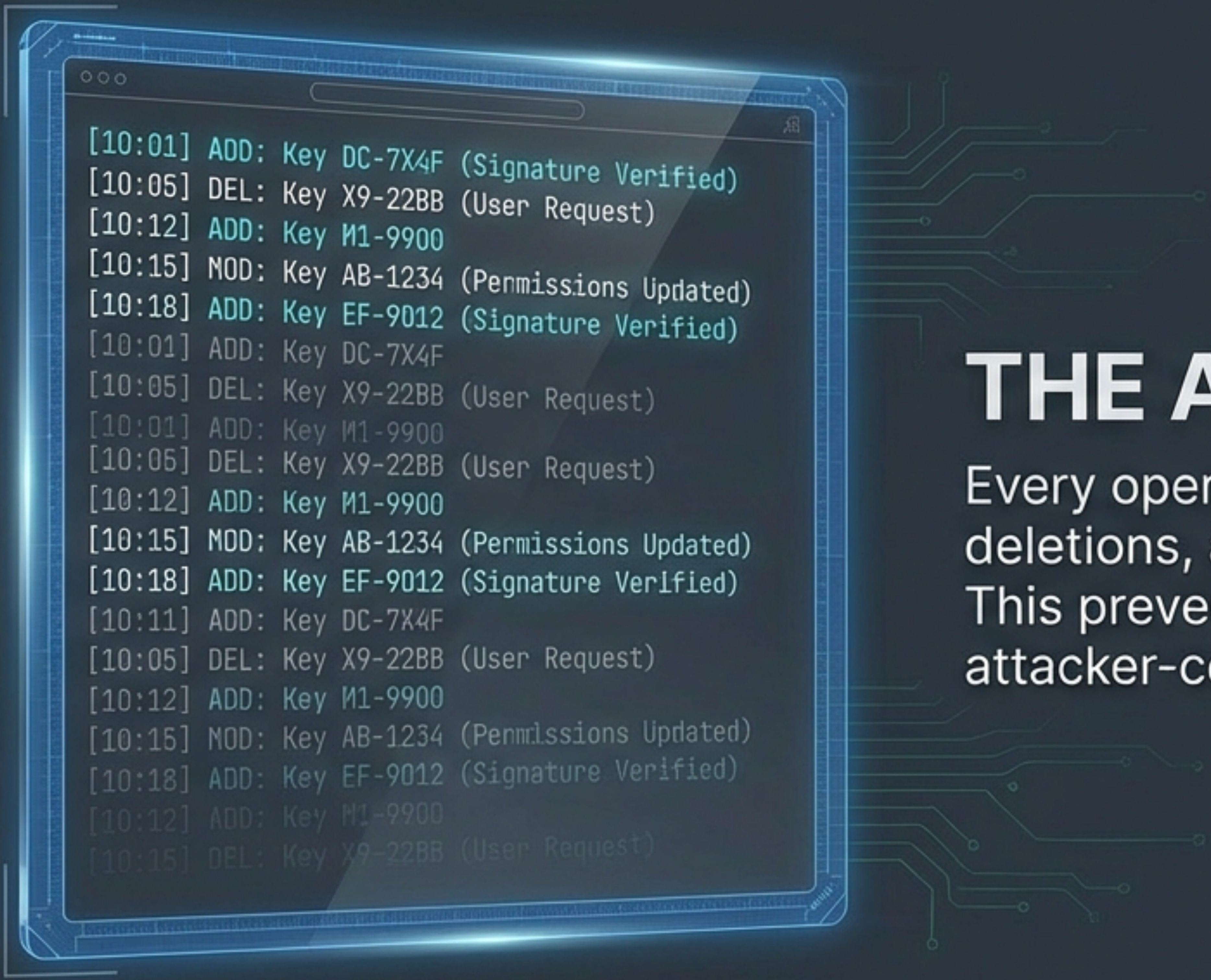
FEATURE 2: PUBLIC KEY REGISTRY

A "Phonebook" for
Public Keys.

Deployment Visibility:

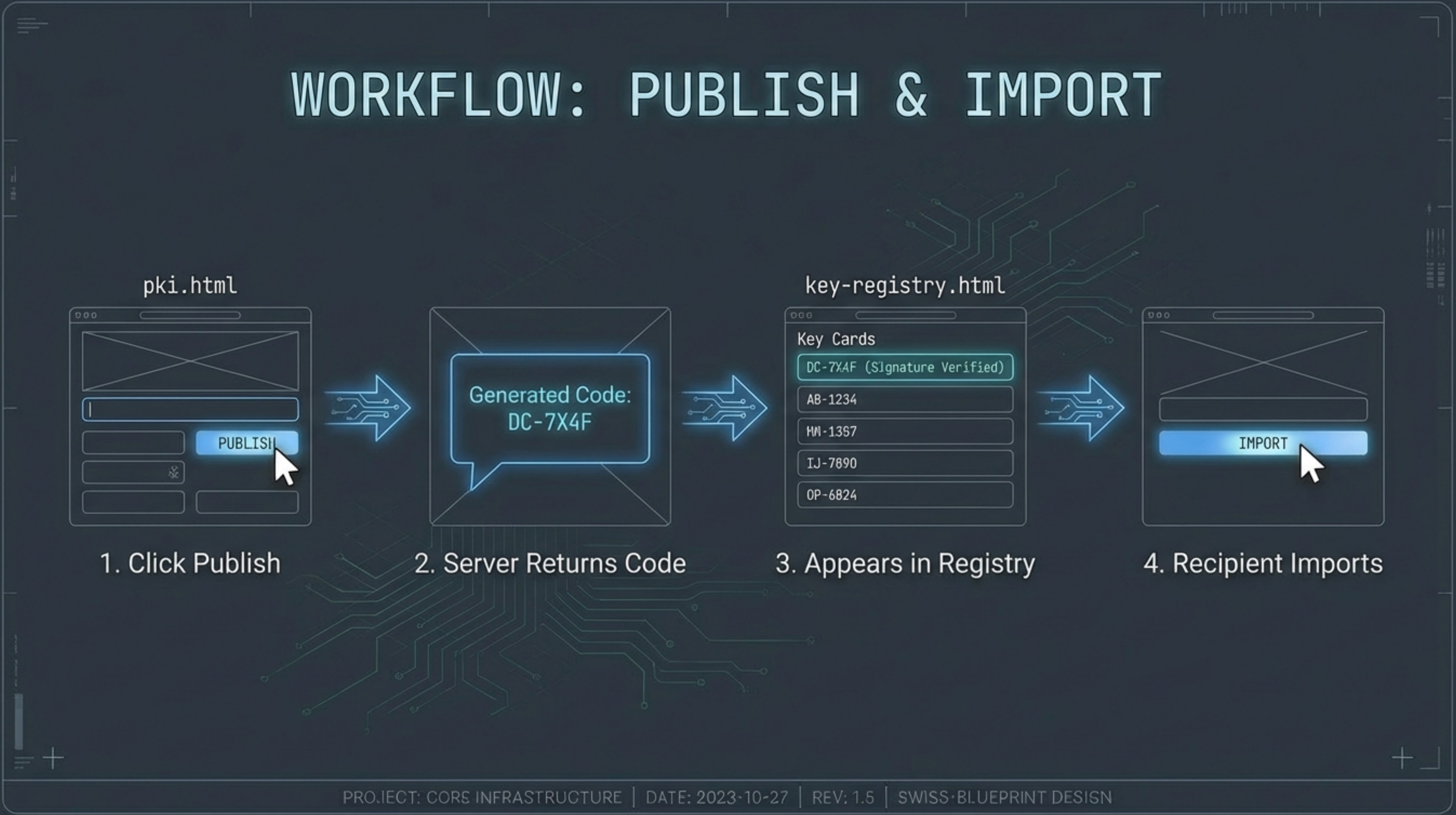
- Private Instance: Authorised users only.
- Public Instance: Open access.
- Analogous to keys.openpgp.org

TRUST & TRANSPARENCY



THE APPEND-ONLY LOG

Every operation is recorded. Additions, deletions, and changes are visible to all users. This prevents the server from silently serving attacker-controlled keys.


WORKFLOW: PUBLISH & IMPORT
pki.html
PUBLISH
Generated Code:
DC-7X4F
key-registry.html
Key Cards
DC-7X4F (Signature Verified)
AB-1234
MN-1357
IJ-7890
OP-6824
IMPORT
1. Click Publish
2. Server Returns Code
3. Appears in Registry
4. Recipient Imports
PROJECT: CORS INFRASTRUCTURE | DATE: 2023-10-27 | REV: 1.5 | SWISS·BLUEPRINT DESISN

TECHNICAL SPECS: API ENDPOINTS

METHOD	ENDPOINT	PURPOSE
POST	/api/keys	Publish public key -> Returns lookup code
GET	/api/keys/{code}	Retrieve public key by lookup code
DELETE	/api/keys/{code}	Unpublish a key (Auth required)
GET	/api/keys	List all published keys (Registry)

NOTE: All operations must interface with the Transparency Log.

STORAGE & FILE STRUCTURE

- /admin/
  - pki.html
  - key-lookup.html
  - key-registry.html
  - js/
    - pki-common.js **(EXTRACTED UTILS)**

Server Storage

- **Key:** Token (DC-7X4F)
- **Value:** Encrypted Blob
- **Log:** Append-only History

Client Storage - IndexedDB

- **Key:** Friendly Name (Dinis Cruz)
- **Value:** Token (DC-7X4F)

ACCEPTANCE CRITERIA

01 ☐ Publish from `pki.html` returns lookup code.
02 ☐ Lookup by code displays key.
03 ☐ Import from lookup adds to contacts.
04 ☐ Registry page lists all keys.
05 ☐ Import from registry works.
06 ☐ Code format is 6–8 alphanumeric chars.
07 ☐ **CRITICAL: No plaintext names on server.**
08 ☐ Transparency log records all ops.
09 ☐ Unpublish works (returns 404).
10 ☐ QR code generation for lookup code works.

IMMEDIATE EXECUTION STRATEGY

PRIORITY 1: Implement ‘Hybrid’ token logic.

PRIORITY 2: Build API endpoints.

PRIORITY 3: Construct UI views (Lookup & Registry).

STATUS: GO